

EGBI 100

Biomedical Engineering In the Real World

Registration Check

Before we begin the course, please make sure you are **officially registered**.

Please note:

- Attendance will be checked against the **official enrollment list** during the first class.
- If you are attending the class **but your name is not on the class list**, or if you have any registration issues, **please inform me immediately**.
- The **Add/Drop period is 3–17 August 2026**. Please resolve any registration issues within this period.

Course Information

- Credit: 1 (1-0-2) Pre-requisite: None
- Time: **Tuesday 5.40-6.40 PM**
- Place: **R238 Faculty of Engineering Building 1, Mahidol University**

- **Course description:**

Historical development and survey of major areas comprising biomedical engineering fields of activity; research, development, and design for biomedical problems, diagnosis of diseases, and therapeutic applications

Course coordinator



Aj. Petch

Dr. Pinunta Nittayacharn

Email: Pinunta.nit@mahidol.ac.th

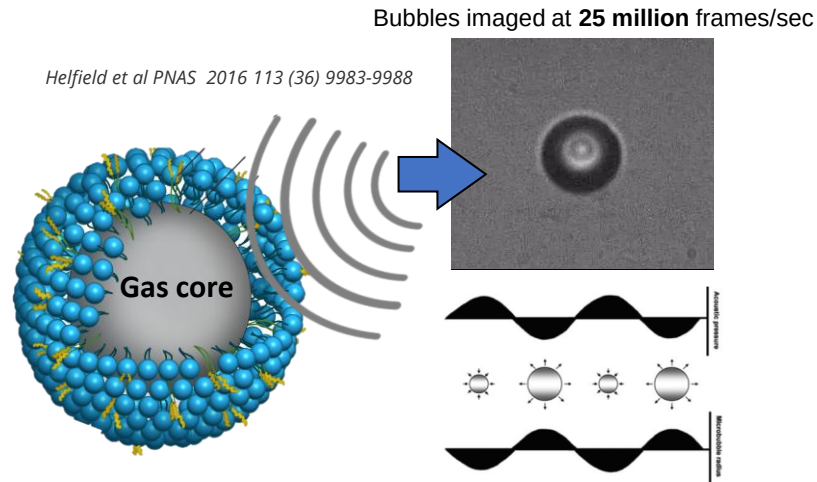
- **Office:** Room 6455, 4th Floor, Department of Biomedical Engineering, Engineering Building 3
- **Office Hour:** By appointment via email

Tissue Engineering and Drug Delivery Systems

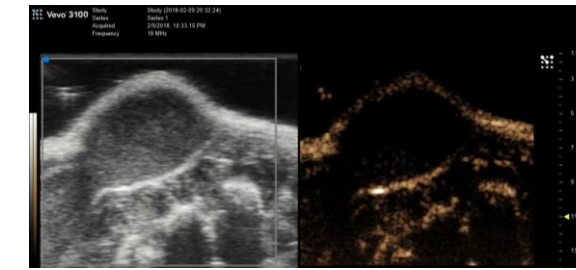


Pinunta Nittayacharn, Ph.D.

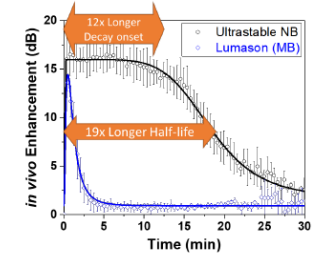
Nanobubbles for ultrasound imaging and drug delivery



Highly ultrasound visible (flank tumor)

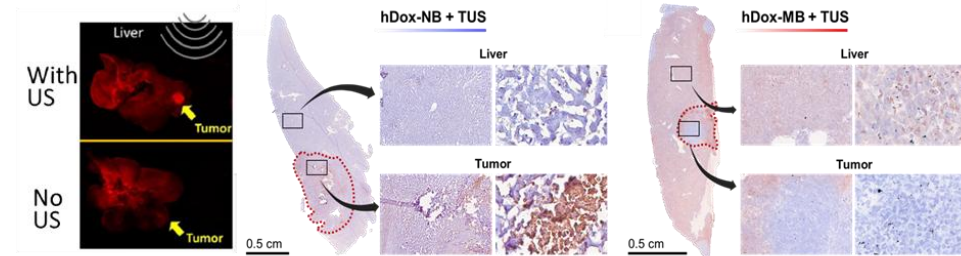


Nanoscale, 2019, 11, 15647-15658



20x longer lasting compared to clinical MBs

Increase tumor specificity with hDox-NB and US stimulation



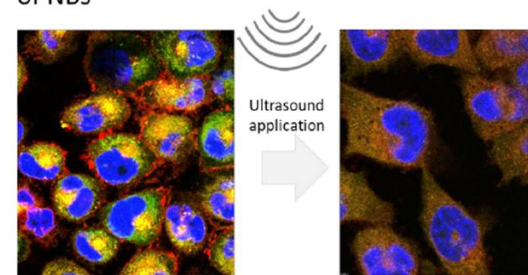
Journal of Controlled Release 2023, 367, 135-147



Research Interests

- Ultrasound contrast agents
- Image-guided drug/gene delivery
- Cancer detection and therapy
- Nanomedicine
- Molecular imaging

Endosomal escape using ultrasound stimulation of NBs



Nuclei, nanobubbles, lysosomes
nanobubble/lysosome co-localization

Nanotheranostics. 2022, 6 (3), 270

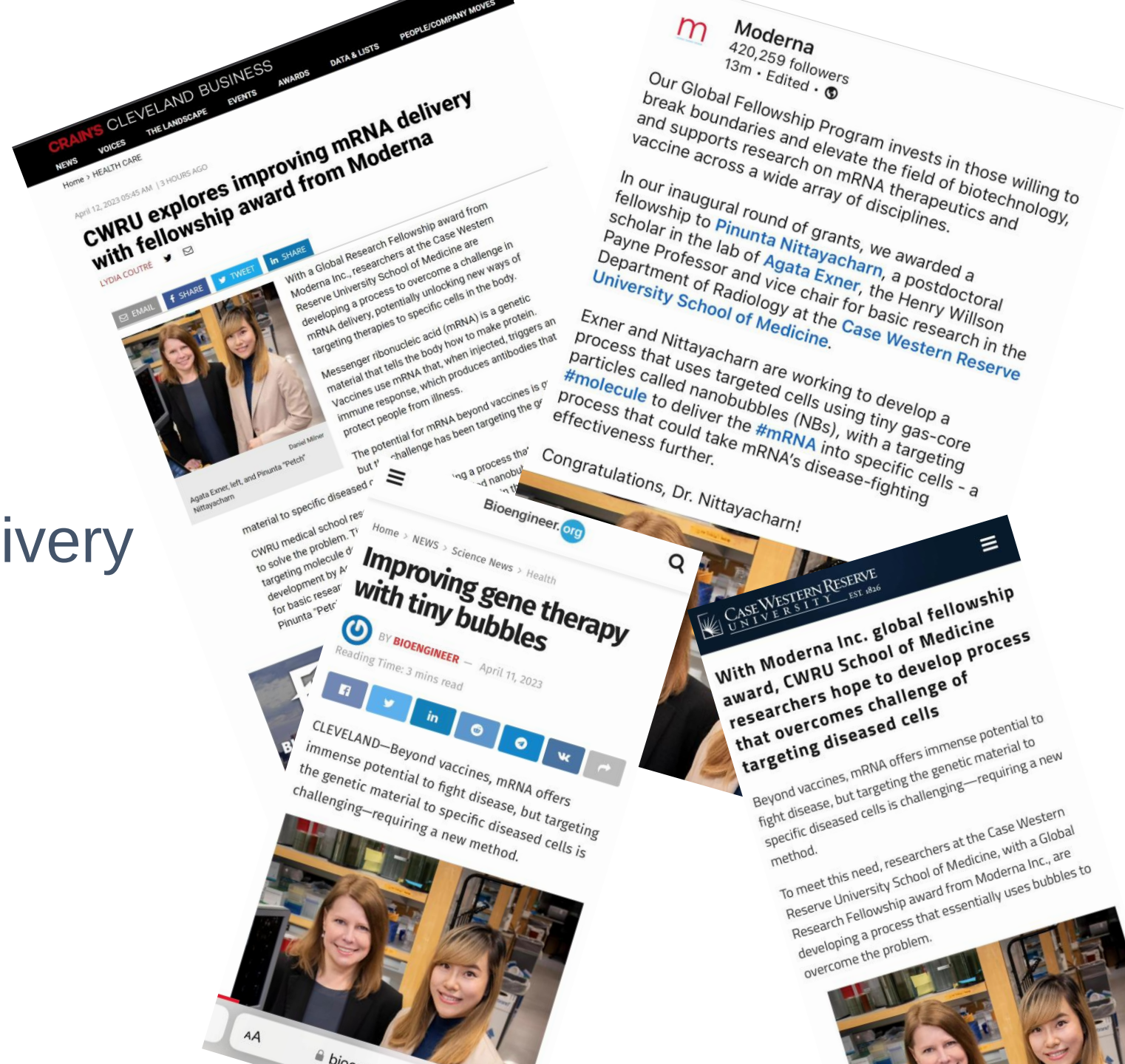
moderna



CASE
WESTERN
RESERVE
UNIVERSITY

Ultrasound-Mediated, Targeted Drug/Gene Delivery Using Gas-Core Lipid Nanoparticles

Ph.D. Dissertation and Postdoctoral Research



BME Faculty Members

<https://www.eg.mahidol.ac.th/dept/egbe/>



Jackrit Suthakorn, Ph.D.
Leader, Medical Robotics
Research Interests - Robot-
Assisted Surgery, Surgical
Navigations



Warakorn Charoensuk, Ph.D.
Research Interests – Human Motion
Analysis and Rehabilitation Technology
Mixed reality- (MR) based health-related
physical fitness assessment system
Prostheses and Orthoses



Norased Nasongkla, Ph.D.
Leader, Tissue Engineering and Drug
Delivery Systems Research Track



Yodchanan Wongsawat, Ph.D.
Leader, Biosignal and Image Processing
Brain Computer Interface



Chamras Promptmas, Ph.D.
Director, Biosensors and
Micro Total Analysis System
Laboratory



Panrasee Ritthipravat, Ph.D.
Director, AI in Medicine Laboratory



Songpol Ongwattanakul, Ph.D.
Director, Biomedical Cloud Computing
Laboratory



Theeraporn Rubcumintara, Ph.D.
Material Engineering, Biomaterial



Phornphop Naiyanetr, Ph.D.
Leader, Rehabilitation
Engineering and Artificial
Organ Research Track



Pracha Yambangyang, Ph.D.
Director, Biomedical instrumentation
Laboratory
Biomedical Instrumentation,
Clinical Engineering
Hospital Engineering



Benchaporn Lertanantawong, Ph.D.
Nanobiosensors for biomedical
application, DNA self-assembly,
Electrochemistry, Nanomaterials for
sensing application,



Titipat Achakulvisut, Ph.D.
Applied Machine Learning,
Natural Language Processing,
Metascience



Soontorn Oraintara, Ph.D.
Biomedical signal and image
processing,
Modeling of biomedical signals
and images



Jetsada Arnin, Ph.D.
Public Relation Coordinator
Neuro-Rehabilitation,
Wireless Embedded Systems



Pronpat Athamanolap, Ph.D.
Molecular Diagnostics, Microfluidics,
Point-of-Care Diagnostics,
Infectious Diseases, Epigenetics



Chalaisorn Thanapongpibul, Ph.D.
Nanotherapeutics and nanomaterials



Shen Treratanakulchai, Ph.D.
Medical Robotics,
Soft Robotics,
Manipulator,
Mechatronics



Panya Kaimuk, M.D.
Sports Science, Golf,
Orthopedic Surgery



Choladawan Moonjaita, M.Eng.
Robot Design



Yuttana Itsarachaiyot, Ph.D.
Medical Robotics
Continuum Robots
Human Machine Interface

Latest Emerging Technologies in Healthcare in 2025

1. Artificial Intelligence in Healthcare
2. Integrating Data and Predictive Analysis
3. Nanomedicine
4. Augmented & Virtual Reality in Healthcare
5. Technology in Mental Health
6. Remote Patient Monitoring & Virtual Care
7. Digital Twins for Health
8. Wearables in Healthcare
9. Organ Care Technology & Bioprinting
10. Healthcare Data Management

Course contents (may be adjusted during the semester)

Week	Date (Tuesday 5.40- 6.40 PM)	Topics	Lecturer
1	Aug 04	Course Overview & Nanomedicine	Dr. Pinunta
2	Aug 11	Regenerative medicine	Dr. Chalaisorn
3	Aug 18	Molecular diagnostics	Dr. Pornpat
4	Aug 25	Biosensors	Assoc. Prof. Dr. Benchaporn
5	Sep 01	Drug Delivery System	Assoc. Prof. Dr. Norased
6	Sep 08	Medical Robotics: Rehabilitation, Hospital Service and Logistic	Assoc. Prof. Jackrit
7	Sep 15	TBD	Dr. Yuttana
8	Sep 22	Soft Robotics for Medicine	Dr. Shen

Course contents (may be adjusted during the semester)

Week	Date (Tuesday 5.40- 6.40 PM)	Topics	Lecturer
9	Oct 20	Cloud Computing in Healthcare	Dr. Songpol
10	Oct 27	3D imaging for healthcare	Dr. Jetsada
11	Nov 03	Artificial intelligence in medicine	Assoc. Prof. Dr. Panrasee
12	Nov 10	Data Science in Healthcare and Medicine	Dr. Titipat
13	Nov 17	Rehabilitation Technology	Asst. Prof. Dr. Warakorn
14	Nov 24	Biomedical Instrumentation	Dr. Pracha

**There are no class on Oct 6 and 13.

Communication tools-Online platforms

- I would like to make sure we can communicate well and very efficiently through various useful online platforms.
- Email
- Line Group
- Google Classroom
 - Class Code: klrxkicn



Grading policy (O/S/U)

• Attendance	15%
• Assignments/Pop-Quiz	40%
• Term project-report	10%
• Term project-presentation	35%

U < 60%

S ≥ 60%

O ≥ 90%

Class attendance



Attendance will be recorded weekly.

- Each week, I will ask either the class representative or the lecturer to **randomly take a photo of the entire class**.
- Attendance will be determined by **counting the students present in the photo**.
- Please ensure your face is visible in the class photo so your attendance can be verified.

Please Note:

- ☒ **Arrive on time.** Late arrivals may not be included in the attendance record.
- If you are unable to attend a class, **please email the lecturer for that week before the class** (or as soon as possible) and **CC me** in the email.
- Attendance contributes to your participation in this course, so please attend every session whenever possible.

Class attendance

Must attend at least 80% (~12 lectures) to pass the class.

We consider **attendance mandatory**.

If you are going to miss class, notify us in advance **at least 1-2 days in advance** with valid reason (unless it's emergency).

via email: Pinunta.nit@mahidol.ac.th

Subject: EGB100-(student ID)-" the purpose of your email", for example: EGB100-69xxxxx-Absence from classes

Content: Introduce yourself and explain the reason why you will be (/were) unable to attend the class.

Assignments – individual

Assignment 1. Lecture Reflection (30%)

- **Objective:** To encourage students to critically reflect on the lecture content and relate it to biomedical engineering applications.
- **Instructions:**
- Select **two (2)** lectures from the course that you found the most interesting or impactful.
- For **each lecture**, prepare a reflection (**maximum 1 page**, approximately **200–300 words**) that includes:
 - **Lecture Title**
 - **Key Takeaways** (3–5 bullet points)
 - **Reflection**
 - Why did you choose this lecture?
 - What was the most important concept you learned?
 - How can this knowledge be applied to biomedical engineering, healthcare, or future research?
 - **One Question**
 - Identify one question or topic you would like to explore further.

Assignments – individual

Assignment 2. Emerging Technology Exploration (70%)

- **Objective:** To encourage students to explore and critically evaluate an emerging healthcare technology beyond the topics covered in class.
- **Instructions**
- Conduct research on a **current and emerging healthcare technology** in the field of biomedical engineering **that has not been covered in this course.**
- Your selected technology may be based on:
 - **A recent original research article** (published within the last **5 years**), or
 - **A commercialized product or technology** that demonstrates an innovative biomedical engineering application.
- **Note:** The selected technology should represent a **novel or emerging innovation**, rather than a well-established or conventional technology.

Assignments – individual

Prepare a **2-page report** (including references) that includes:

1. Technology Overview

- Name of the technology
- Research article or commercial product information (citation or product source)
- Healthcare problem or unmet clinical need addressed

2. Technology Description

- How does the technology work?
- What makes it innovative compared with existing technologies?

3. Current Applications

- Current or potential clinical/research applications
- Advantages and limitations

Assignments – individual

Prepare a **2-page report** (including references) that includes:

4. Critical Evaluation

Discuss:

- Strengths of the technology
- Challenges or limitations for clinical translation or commercialization
- Future improvements or research directions

5. Personal Reflection

- Why did you choose this technology?
- What did you learn from exploring it?
- How does it relate to concepts introduced in this course?

6. References

- Include at least **one primary research article** and any additional references or product information used.

Grading Rubric

Assignment 1 – Lecture Reflection (30%)

- Summary of key concepts: 30%
- Critical reflection and application: 50%
- Organization and writing quality: 20%

Assignment 2 – Emerging Technology Exploration (70%)

- Technology selection and background: 30%
- Critical analysis: 40%
- Personal reflection: 20%
- Organization and references: 10%

Assignments – individual

- **The deadline is on Dec 1st, 2026, at 11:59 PM .**
- Submission is in Google classroom. **Late submission will not be accepted.**
- The **file name must be ID_Firstname** e.g. 69xxxxxx_Nancy
- Plagiarism and copying are prohibited. You will get 0 point.
- The assignment must be taken completely alone. It is an open-book, open-note. You may use any publicly available material you want, including books, research articles, etc.

Term Project – Group project 3-5 ppl/group

Group Term Project: Innovation in Biomedical Engineering

- **Objective:** To develop an innovative biomedical engineering solution that addresses an unmet healthcare need by integrating scientific knowledge, engineering principles, and creativity.
- **Project Description:** Each group will identify **a healthcare challenge, disease, or unmet clinical need** and propose a **novel biomedical engineering solution**.
- The proposed innovation may include:
 - A new medical device
 - A diagnostic technology
 - A drug delivery system
 - A biomaterial or tissue engineering approach
 - An AI or digital health solution
 - Any other innovative biomedical engineering technology
- Your proposal should demonstrate:
 - The healthcare problem being addressed
 - The scientific and engineering principles behind the innovation
 - How the proposed solution improves upon existing technologies
 - Potential benefits, limitations, and future applications

Term Project – Group project 3-5 ppl/group

Project Deliverables

1. Short Report (5 pages, exclude title page)

Include:

- Title page: Detail each member's contribute on in your report's appendix
- Background and clinical need
- Description of the proposed innovation
- Scientific rationale
- Expected advantages and potential limitations
- References

2. Short Video Pitch (5 minutes)

Present your innovation in a clear and engaging way, including:

- The healthcare challenge
- Your proposed solution
- How it works
- Why it is innovative
- Potential impact on healthcare

Creativity is encouraged! You may use animations, sketches, prototypes, demonstrations, or AI-generated illustrations to communicate your idea effectively.

Term Project – Grading Rubric

Report (10%)

- Background & clinical need (20%)
- Innovation & scientific rationale (50%)
- Organization & references (30%)

Presentation (35%)

- Content & Organization (30%) – Clear, logical, and well-structured presentation.
- Delivery & Communication (30%) – Confident, clear, and engaging delivery.
- Visual Aids (20%) – Effective and professional use of slides, figures, or multimedia.
- Teamwork (10%) – Balanced participation and smooth transitions between speakers.
- Time Management (10%) – Presentation stays within the assigned time limit

Term project

Submission Guidelines:

- Report: Submit as a PDF through the course's google classroom.
- Video Presentation: Upload to a platform like YouTube (can be unlisted) or another preferred platform and **share the link** through the course's google classroom.

File name: e.g., **Team name_Report/ Team name_VDO**

- Only the **group representative** should submit the project files. Please upload **both** the **short report** and the **video presentation** in a single submission on behalf of all group members.

Due Date: Dec 1st, 2026, at 11:59 PM.

What should you do in class?

- Take note (some speaker will provide slide, some may not. So do not rely on slides)
- Ask questions if you do not understand (or anything related to the talk)



We, BART LAB Rescue Robotics Team are at the collapsed 6-story-building site in Pathumthani Klong...



Brain-Computer Interface Race (BCI)

Rank	Team	Score	Time	Play Video
1	MANOJ KULDEB	100%	1:10:00	Play Video
2	MANOJ KULDEB	100%	1:10:00	Play Video
3	MANOJ KULDEB	100%	1:10:00	Play Video
4	MANOJ KULDEB	100%	1:10:00	Play Video
5	MANOJ KULDEB	100%	1:10:00	Play Video

Minister of Industry,
Thailand



ARTICLE

<https://doi.org/10.1038/s41467-021-21121-7> OPEN

Rapid electrochemical detection of coronavirus SARS-CoV-2

Thanyarat Chaibun¹, Jiratchaya Puenpa², Tatchanun Ngamdee³, Nimaradee Boonapatcharoen⁴, Pornpat Athamanolap¹, Anthony Peter O'Mullane⁵, Sompong Vongpunsawad², Yong Poovorawan², Su Yin Lee^{6,7} & Benchaporn Lertanantawong¹✉

Check for updates



ซีเอสอาร์-เอชอาร์

“นักวิจัยไทย” ก้องโลก ใช้นวัตกรรมตรวจโควิด

วันที่ 17 กุมภาพันธ์ 2564 - 09:17 น.

เทคโนโลยีส์

การคิดโควิด-19 รอบใหม่ หางาน สร้างเงิน คนค้าขายออนไลน์ ข่าวเดลิเวียร์ สถานีเดลิเวียร์ บทความ

หน้าแรก ข่าวเดลิเวียร์ / กรม-จราจร

เจ๋ง! วารสารวิทยาศาสตร์ชั้นนำของโลก เผยผลงานนักวิจัยวิศวกรรมศาสตร์ มมหิดล
พัฒนาชุดตรวจโควิด-19 ด้วยเซ็นเซอร์ชีวภาพแม่นยำ 100%

Nature Communications วารสารวิทยาศาสตร์ชั้นนำของโลก เผยผลงานนักวิจัยวิศวกรรมศาสตร์ มมหิดล
พัฒนาชุดตรวจโควิด-19 ด้วยเซ็นเซอร์ชีวภาพแม่นยำ 100%

พุธที่ 10 กุมภาพันธ์ 2564 เวลา 17:15 น.

